

● MEDIUM

● ALGEBRA

● ANSWER KEY

# Systems of two linear equations in two variables

01

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**Q1****B****B. 5**

Choice B is correct. Adding the first equation to the second equation in the given system yields  $x + 2y + x - 2y = 6 + 4$ , or  $x + x + 2y - 2y = 10$ . Combining like terms in this equation yields  $2x = 10$ . Dividing both sides of this equation by 2 yields  $x = 5$ . Thus, the value of  $x$  is 5.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of  $2x$ , not  $x$ .

**Q2****D****D. 4.34**

Choice D is correct. Let  $l$  represent the cost, in dollars, of 1 loaf of bread, and let  $d$  represent the cost, in dollars, of 1 dozen eggs. It's given that the first customer paid 12.45 dollars for 1 loaf of bread and 2 dozen eggs. Therefore, the first customer's purchase can be represented by the equation  $l + 2d = 12.45$ . It's also given that the second customer paid 19.42 dollars for 4 loaves of bread and 1 dozen eggs. Therefore, the second customer's purchase can be represented by the equation  $4l + d = 19.42$ . The equations  $l + 2d = 12.45$  and  $4l + d = 19.42$  form a system of linear equations, which can be solved by elimination to find the value of  $d$ . Multiplying the first equation in the system by  $-4$  yields  $-4l - 8d = -49.8$ . Adding  $-4l - 8d = -49.8$  to the second equation,  $4l + d = 19.42$ , yields  $-4l + 4l + -8d + d = -49.8 + 19.42$ , which is equivalent to  $-7d = -30.38$ . Dividing both sides of this equation by  $-7$  yields  $d = 4.34$ . Therefore, the cost, in dollars, of 1 dozen eggs is 4.34.

Choice A is incorrect. This is the cost, in dollars, of 1 loaf of bread.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

**Q3****B****B. 1**

Choice B is correct. Each given equation is written in slope-intercept form,  $y = mx + b$ , where  $m$  is the slope and  $b$  is the  $y$ -intercept of the graph of the equation in the  $xy$ -plane. The graphs of two lines that have different slopes will intersect at exactly one point. The graph of the first equation is a line with slope 1. The graph of the second equation is a line with slope 8. Since the graphs are lines with different slopes, they will intersect at exactly one point.

Choice A is incorrect because two graphs of linear equations have 0 intersection points only if they are parallel and therefore have the same slope.

Choice C is incorrect because two graphs of linear equations in the  $xy$ -plane can have only 0, 1, or infinitely many points of intersection.

Choice D is incorrect because two graphs of linear equations in the  $xy$ -plane can have only 0, 1, or infinitely many points of intersection.

**Q4****C****C. 9**

Choice C is correct. Since the graph of  $y = x + 3$  intersects the graph of  $y = 2x - 6$  at the point  $(a, b)$ , the ordered pair  $(a, b)$  is the solution to the system of linear equations consisting of  $y = x + 3$  and  $y = 2x - 6$ , and the value of  $a$  is the value of  $x$  in the solution of this system. Since both  $x + 3$  and  $2x - 6$  are equal to  $y$ , it follows that  $x + 3 = 2x - 6$ . Subtracting  $x$  from and adding 6 to both sides of the equation yields  $9 = x$ . Therefore, the value of  $a$  is 9.

Choices A and B are incorrect and may result from a calculation or conceptual error in solving the system of equations consisting of  $y = x + 3$  and  $y = 2x - 6$ . Choice D is incorrect. This is the value of  $b$ , not  $a$ .

**Q5****A**

A.  $y = 0$   
 $x = \frac{3}{2}$

Choice A is correct. The solution to a system of equations is the coordinates of the intersection point of the graphs of the equations in the  $xy$ -plane. Based on the graph, the solution to the given system of equations is best approximated as  $(\frac{3}{2}, 0)$ . In the  $xy$ -plane, the graph of  $y = 0$  is a horizontal line on which every  $y$ -coordinate is 0, and the graph of  $x = \frac{3}{2}$  is a vertical line on which every  $x$ -coordinate is  $\frac{3}{2}$ . These graphs intersect at the point  $(\frac{3}{2}, 0)$ . Therefore, the system of equations in choice A has the same solution as the given system.

Choices B, C, and D are incorrect. If graphed in the  $xy$ -plane, these choices would intersect at the points  $(0, \frac{3}{2})$ ,  $(1, 0)$ , and  $(0, 1)$ , respectively, not  $(\frac{3}{2}, 0)$ .

**Q6****A**

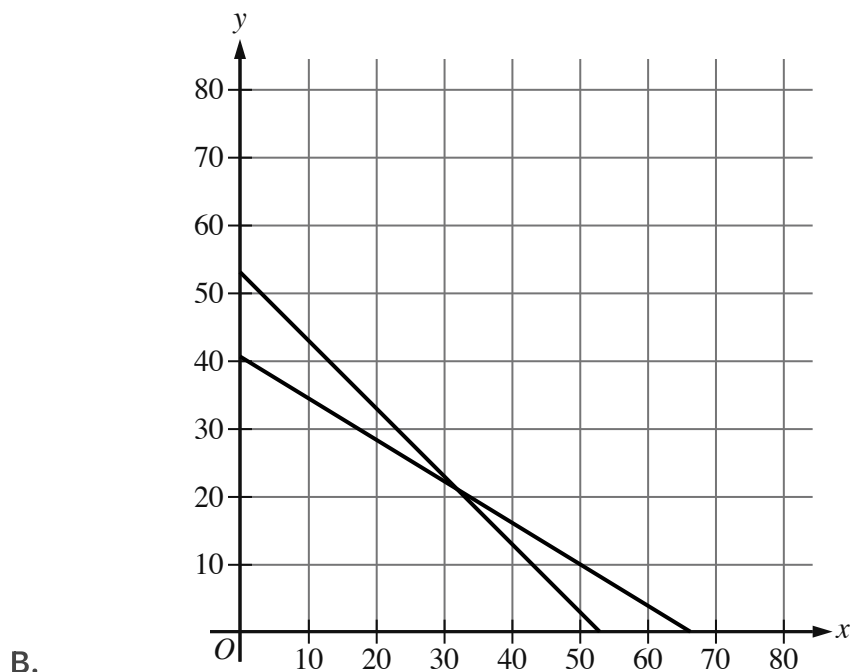
A. -2

Choice A is correct. The given system of linear equations can be solved by the elimination method. Multiplying each side of the second equation in the given system by 3 yields  $2x + 2y(3) = 103$ , or  $6x + 6y = 30$ . Subtracting this equation from the first equation in the given system yields  $6x + 7y - 6x + 6y = 28 - 30$ , which is equivalent to  $6x - 6x + 7y - 6y = 28 - 30$ , or  $y = -2$ .

Choice B is incorrect. This is the value of  $x$ , not the value of  $y$ .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

**Q7****B**

- For the first line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following points:
    - (0 comma 53)
    - (53 comma 0)
- For the second line in the system:
  - The line slants gradually down from left to right.
  - The line passes through the following approximate points:
    - (0 comma 41)
    - (66 comma 0)
- The 2 lines intersect at 1 point.

Choice B is correct. The intercepts of the graph of each equation can be found by substituting 0 for one variable at a time. For the equation  $x + y = 53$ , substituting 0 for  $y$  yields  $x = 53$ , so the  $x$ -intercept of the graph of the equation is  $(53, 0)$ . Substituting 0 for  $x$  in this equation yields  $y = 53$ , so the  $y$ -intercept of the graph of the equation is  $(0, 53)$ . Thus, the graph of the equation  $x + y = 53$  has intercepts at  $(53, 0)$  and  $(0, 53)$ . For the equation  $11x + 18y = 730$ , substituting 0 for  $y$  yields  $x = \frac{730}{11}$ , so the  $x$ -intercept of the graph of the equation is approximately  $(66, 0)$ . Substituting 0 for  $x$  in this equation yields  $y = \frac{730}{18}$ , so the  $y$ -intercept of

the graph of the equation is approximately  $(0, 41)$ . Thus, the graph of the equation  $11x + 18y = 730$  has intercepts at approximately  $(66, 0)$  and  $(0, 41)$ . Choice B is a graph that shows one line with intercepts at  $(0, 53)$  and  $(53, 0)$  and another line with intercepts at approximately  $(0, 41)$  and  $(66, 0)$ . Thus, choice B represents this situation.

Choice A is incorrect. This graph represents the equations  $y = x$  and  $11x + 18y = 730$ .

Choice C is incorrect. This graph represents the equations  $x + y = 53$  and  $-11x + 18y = 730$ .

Choice D is incorrect. This graph represents the equations  $y = 53$  and  $-11x + 18y = 730$ .

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**Q8**

40

The correct answer is 40. It's given in the first equation of the system that  $y = -2x$ . Substituting  $-2x$  for  $y$  in the second equation of the system yields  $3x + -2x = 40$ . Combining like terms on the left-hand side of this equation yields  $x = 40$ . Therefore, the value of  $x$  is 40.

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**Q9**

1

The correct answer is 1. Subtracting the second equation from the first equation in the given system of equations yields  $3x - 3x + 6 - 4 = 4y - 2y$ , which is equivalent to  $0 + 2 = 2y$ , or  $2 = 2y$ . Dividing each side of this equation by 2 yields  $1 = y$ .

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**Q10**

24

The correct answer is 24. The given system of equations can be solved by the substitution method. The first equation in the given system of equations is  $y = \frac{1}{3}x - 14$ . Substituting  $\frac{1}{3}x - 14$  for  $y$  in the second equation in the given system yields  $\frac{1}{3}x - 14 = -x + 18$ . Adding 14 to both sides of this equation yields  $\frac{1}{3}x = -x + 32$ . Adding  $x$  to both sides of this equation yields  $\frac{4}{3}x = 32$ . Multiplying both sides of this equation by  $\frac{3}{4}$  yields  $x = 24$ .

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#### Notes

Accept equivalent forms (e.g.  $0.25 \equiv 1/4$ ). For grid-in items, decimal and fraction answers are both valid.

